

SYED ISLAM

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LLM infrastructure and applied ML engineer — production RAG, evaluation harnesses, and large-scale data pipelines.

EDUCATION

University of Texas at Austin Sep 2025 – May 2027
MS in Artificial Intelligence · GPA 3.6 Austin, TX
Coursework: Machine Learning, Deep Learning, Distributed Systems, Databases, Optimization.

University of California, Riverside Sep 2023 – Jun 2025
BS in Computer Science with Business Applications · GPA 3.8 Riverside, CA

EXPERIENCE

Carboncopies Foundation Jun 2026 – Present
LLM Infrastructure Intern Remote

- Architected CarbonGPT, a TypeScript/TanStack Start app unifying the React SPA with an OpenAI-compatible chat API over a LangChain.js RAG agent (llama.cpp, ChromaDB, vLLM on a dedicated GPU VM).
- Designed auth and state: Google OAuth httpOnly sessions, owner-scoped history in Postgres, guest-only history in IndexedDB, cross-user isolation tests.
- Delivered Phase 1 — RAG chat with live retrieval and citations, replacing a fragile Flowise UI — and authored the 8-task Phase 2 plan for auth, persistence, and negative security tests.

Algoverse AI Research Program May 2026 – Present
AI Research Fellow Remote

- Investigating how vision-language models encode emotion and appraisal signals across text and images, probing pleasantness, predictability, and agency on crowd-enVENT and EMOTIC.
- Built an interpretability pipeline over Gemma-3 multimodal with TransformerLens — ridge probes, unique-effect vectors, and targeted steering to test whether shifting appraisal directions moves emotion label distributions in theory-predicted directions.
- Working in a 4-person team under weekly mentorship toward a NeurIPS/EMNLP-ready manuscript.

Carboncopies Foundation Jun 2025 – Aug 2025
Software Engineer Intern Remote

- Engineered a role-based API server (contributor/admin) for secure CLI operations.
- Migrated the API server from Restbed to Oat++, improving request throughput and maintainability.
- Designed an export API for connectome data (neuron types, 3D locations, synapse details).

PROJECTS

High-Risk LLM Clinical Decision Support colab.com/LLM-Clinical

- Built a RAG pipeline over 8,975 Chinese medicine terms (56 groups) with sub-10 ms retrieval per clinician query (sentence transformers, FAISS, Docker).
- Benchmarked four LLMs (llama3.2, deepseek-r1, gpt-oss, gemma) with automated F1/precision/recall harnesses.

Chicago Crime Data Analysis github.com/ChicagoCrimeDataAnalysis

- Led a 5-person team processing 14M+ crime records through a CSV→Parquet pipeline that cut storage ~80% and enabled interactive SQL over the full set (Spark, Python, QGIS).
- Built logistic-regression arrest prediction: 0.81/0.79 → 0.83/0.80 as data grew 10K → 100K.

TECHNICAL SKILLS

Languages	Python, SQL, C++, JavaScript/TypeScript
ML & AI	PyTorch, scikit-learn, Transformers, NumPy, Pandas, RAG (LangChain.js, FAISS, ChromaDB, pgvector), vLLM, llama.cpp, TransformerLens, linear/ridge probes, causal tracing
Data & Infra	Apache Spark, Postgres/pgvector, FastAPI, Docker, Kubernetes, REST/OpenAI-style APIs, Google OAuth, CI/CD, Linux, geospatial analysis (QGIS)
Frontend	React, TanStack Start, Tailwind CSS